

Divy Mistry

+91 9104067502 | divymistry11@gmail.com | [linkedin.com/in/divy-mistry](https://www.linkedin.com/in/divy-mistry) | github.com/LostFrequencies11

PROFILE

- Last year Electronics and Communication student, **specializing in embedded systems**, proficient in C & Python, **Controller based Firmware and Hardware design**; Hands-on experience in **Schematic making, PCB prototyping and Firmware designing in various softwares and IDEs**.
- **Independently designed, developed, tested and debugged** embedded C projects; a fast learner with a strong ability to grasp in any environment and collaborate effectively within cross-functional teams on complex systems.

EDUCATION

LJ University	2023-2026	Shree Durga Vidhyalaya	2022
Bachelor of Electronics and communication		Higher Secondary	
SPI (SEM-6) : 7.61 / 10.0		Pr: 48%	

SKILL

- Languages : C, Embedded C, Python, HTML/CSS, JavaScript
- Hardware: Arduino(UNO,Nano,Nano BLE 33 sense), ESP32(Wroom,C3,S3), NodeMCU8266, STM32(C0, H7, F7, F1, F0), Renesas(RA,RX)
- Protocols : SPI, I2C, I2S, UART, USB, CAN(Basic), RS485, Modbus, RS232, Ethernet, TCP/IP, UDP, WiFi, LoRa, BLE, MQTT
- Developer Tools: STMCUBE IDE, ArduinoIDE, Esp-IDF, E2 Studio, AntiGravity, VS Code, Altium, KiCad, EAGLE
- Soft Skills: Team Player, Fast learner, Effective Communicator
- Language: Gujarati (Native), Hindi (Fluent), English (Intermediate)

WORK EXPERIENCE

SWASEMI PVT LTD	Pirana-Piplaj road
Embedded Intern	Nov 2025 - Present

- I functioned as a Hardware & Firmware Intern, gaining comprehensive hands-on experience from product inception to completion.
- I engaged with STM32, Renesas, ESP32 modules, and WCH32 Controllers series firmware, utilizing their respective IDEs, alongside Altium, Eagle, and KiCad for hardware design and verification.
- I also contributed to industrial sensor production and worked with communication protocols such as RS485, Modbus, RS232, and more.
- I gained extensive exposure to IoT technologies, including MQTT, BLE, Wi-Fi, and cellular networks, as well as network-based protocols like TCP and UDP.
- I undertook various mini or side projects focused on ESP32 series controllers.
- During this internship, I also gained experience with SIMCOM and Quectel modules for data communication via MQTT.
- In this internship, I learned how a product is actually made, tested, and sold to industry standards.
- I was also part of the verification and technical purchase department when testing on a high scale and procurement was needed.
- I also went on-site for testing or visiting to check various parameters.

PROJECTS

Plant Care

JAN 2023 - FEB 2023

NodeMCU8266, DHT11 Sensor, Capacitive Soil Moisture Sensor, Relay Module, Arduino IoT Cloud, ArduinoIDE

- I designed a system to monitor a plant's moisture and humidity levels automatically, Watering automatically to maintain its moisture. It is also connected to the IoT cloud (Arduino Cloud), allowing data access from anywhere.
- I use NodeMCU8266 Microcontroller for cloud connectivity, integrating a DHT sensor, a capacitive moisture sensor, and a relay for effective data reading and water management.

Trackerr

JAN 2025 - FEB 2025

Esp32, Arduino nano BLE 33, Lo-Ra sx1278, Neo-6M GPS module, Oled screen, Press Buttons, Buzzer, DHT11 Sensor, ArduinoIDE, KiCAD, Altium

- I made very useful low power and long range device that can send and receive data from different locations.
- In this device, with the help of Arduino Nano 33 (Tx) and ESP32 (Tx & Rx), LoRa SX1278 sent GPS Module, DHT11, and RSSI values to the receiver and printed them on the OLED display.
- To make this project less complex, i converted my circuitry to PCB that i designed on KiCad and Altium.

Parking management with STM32

DEC 2024-DEC 2024

STM32f030f4p6, IR sensor, LCD display, STM32Cube IDE

- With the help of STM32 and IR sensors, I created parking system that displays the slots that are empty and the slots that are full.
- If the slots are full, it will print a message: "THE PARKING IS FULL."
- The display of the LCD 16x2 is connected to the STM32F030F4P6 and configured with the help of the LiquidCrystal library.

Smart Door security

AUG 2023 - SEP 2023

Esp32, HTML, ArduinoIDE, Buzzer, SMTP

- I designed this system to allow users to enter an ID and password on a webpage managed by an ESP32 device and the door will be opened if ID password is correct.
- If the ID or password is entered incorrectly three times, a buzzer will activate, and an email will be sent to the owner via SMTP.

Audio Amplifier

AUG 2023 - SEP 2023

LM386, Aux cable, Speaker, Capacitor, Resistor

- We made a project to amplify the audio that is given through an AUX cable. This circuit takes a stereo input and, after converting it into mono, passes it through an LM386, resulting in a louder output on the speaker.

Fluid Level Detection

JUL 2025 - AUG 2025

Raspberry Pi 3, Python, Colab, TensorflowLite

- We Utilized Python DataFrame along with TensorFlow and its lite version, we train the model based on CNN deep learning algorithms. After converting it into a TensorFlow lite model, we transfer that model to a Raspberry Pi for detection.
- We created our own dataset for training the model using an IP camera.
- To display the output, we capture the third frame from the camera to assess the image level. If the level is below 75% or above 80%, the bottle will be rejected; otherwise, it will be accepted.